ML Assignment 3

UNIVERSITY OF HARTFORD

Multiple Linear Regression

with Advanced Extensions

Building a Multiple Linear Regression Model to predict house price ranges.

Student: John Bennett

Course: CS569 Machine Learning

Fall 2025

October 14, 2025

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1 Objective

In this assignment, a linear regression model based on multiple features was built, evaluated, and improved upon by polynomial features and regularization. Procedures included:

- Created basic linear regression model
- Extended basic model with Polynomial Features
- Applied Ridge Regression for regularization
- Visualized the results and analyzed indiviual feature impacts

2 Data Set Overview and Cleaning

Table 1: Summary Statistics of House Prices

attribute	mean	std	min	25%	50%	75%	max	data_type	num_null	num_unique
Square_Footage	2815.42	1255.51	503.00	1749.50	2862.50	3849.50	4999.00	int64	0	894
Num_Bedrooms	2.99	1.43	1.00	2.00	3.00	4.00	5.00	int64	0	5
Num_Bathrooms	1.97	0.82	1.00	1.00	2.00	3.00	3.00	int64	0	3
Year_Built	1986.55	20.63	1950.00	1969.00	1986.00	2004.25	2022.00	int64	0	73
Lot_Size	2.78	1.30	0.51	1.67	2.81	3.92	4.99	float64	0	1000
Garage_Size	1.02	0.81	0.00	0.00	1.00	2.00	2.00	int64	0	3
Neighborhood_Quality	5.62	2.89	1.00	3.00	6.00	8.00	10.00	int64	0	10
House_Price	618861.02	253568.06	111626.85	401648.23	628267.29	827141.28	1108236.84	float64	0	1000

The data set used for training and testing of the multiple linear regression model consists of 1000 records each with 8 features, representing unique properties and their particular characteristics. The data is very clean with few issues to be addressed. There are no explicit null values. The number of unique values, and the data types are mostly consistent with each other, indicating limited need to impute or transform any of the features at this stage.

One potential exception for cleaning is the "Garage_Size" feature where its minimum value and 25th percentile is both zero. This value is implicitly a null value, indicating that the property does not have a garage. In context of the report, this value indicates that at least 25% of the data records do not have a garage. The other values from this size are "1" and "2", based on the fact that they are the median and max values, respectively, and there are only a total of three unique values in the feature. These values either quantify the number of distinct garages or the number of bays in the garage, but it is not clear which is the case. In either case "0" indicates no garage. This means that "Garage_Size" could be separated into two variables- one that indicates whether the property has a garage and the other that quantifies the number of garages/bays. However, since the current feature unambiguously encodes this information as is, the feature is left in its original state.

All of the features are accurately characterized by its data type with the exception of the "Neighborhood_Quality" attribute whose numeric values are proxies for ordinal quantitative grades of the neighborhood in which each property is found. There is no meaningful magnitude inherent in the values and as such the feature as is will not be effectively interpreted within the linear regression algorithm as is. Because of this, the feature can be converted to a categorical type that will eventually be encoded with rank

in preparation for fitting the linear model. This would ensure that the values will not be treated as interval data types, in spite of the values being numbers. However, unless number-valued ordinal data's values have an inverse rank, it is best to leave it as is. In this case the rank is directly aligned with the set of values the feature can take. The values 1-10 represent the rating of neighborhoods lowest to highest already (kaggle). When creating ordinal data types, the linear regression algorithm will eventually need it to be converted back to numbers to satisfy the needs of the algorithm.

Ultimately cleaning was not required for this dataset.

3 Data Preparation and Model Training

Three different approaches for generating multiple linear regression models were carried out and the results of their individual techniques were compared. This included building a basic linear regression model as well as models that include polynomial transformations of numeric features in order to enhance the performance of the trained model and regularization of model model to increase precision of the model. Evaluative statistics were collected for each model technique to assess the their relative effectiveness. Simple linear models were built that corresponded to the techniques used in for the multiple regression models, providing insight into the amount of variance that each individual feature accounted for in the data.

3.1 Basic Linear Model

The basic linear model utilized all available features in the source data without consideration for the individual contributions of each feature to the overall variance in the data explained by the model. In addition to the evaluative statistics an corresponding simple linear models, feature performance was evaluated to provide insight into what features had the greatest impact on the multiple regression model and how that corresponded to the effectiveness of the simple linear models.

3.2 Polynomial Features Model

Next multiple linear regression models were produced. The numeric features were were transformed into polynomial features that distribute the information contained in a single feature across multiple features. This distribution of information allows for non-linear relationships between features to be captured in the model which is reflected in its predictions. These relationships are established through raising the degree on the values of each independent variable as separate features as well as defining features that include the product between two features. This significantly augments the dimensions to the training and testing data and can potentially cause overfitting but is a reliable method for improving linear models. Fitting the simple models using the same polynomial degrees of the multiple regression models, clarifies how the predictive curve becomes more responsive to patterns in the data.

3.3 Ridge Regression Polynomial Features Model

Finally, multi linear regression models were produced using the ridge regression technique. This is a regularization technique for increasing precision on a module while sometimes introducing bias. Often this bias is very minimal making the techniques ability to increase precision worth the trade off. These models each used the same degrees of polynomial enhancement used in the prior models and can sometimes improve on multi-colinearity that exists between the independent features of linear models. In the context of polynomial features regularization can mitigate some of the multi-colinearity that is introduced by the dimensional expansion of the source data.

4 Results and Discussion

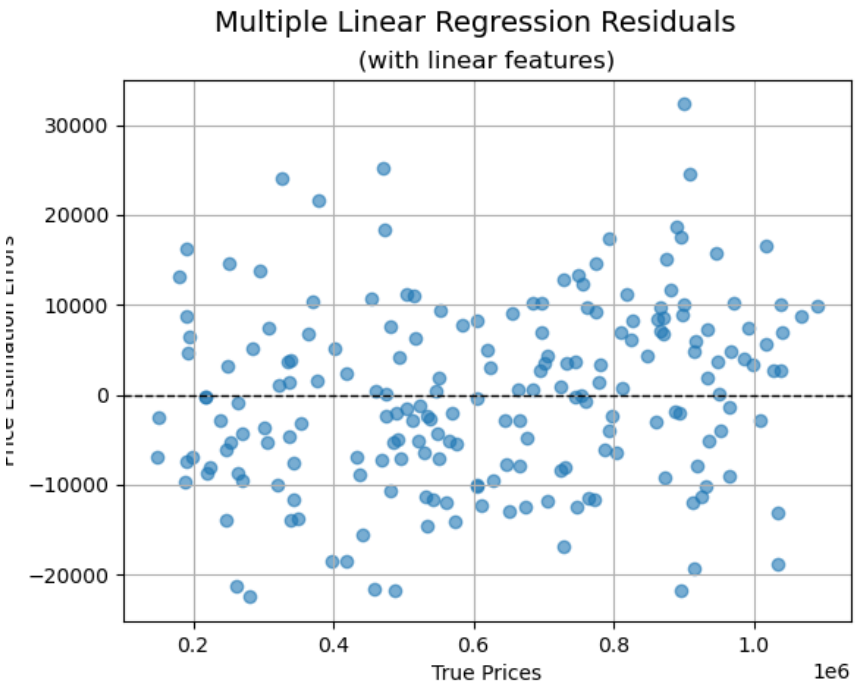
Table 1: Model Comparison

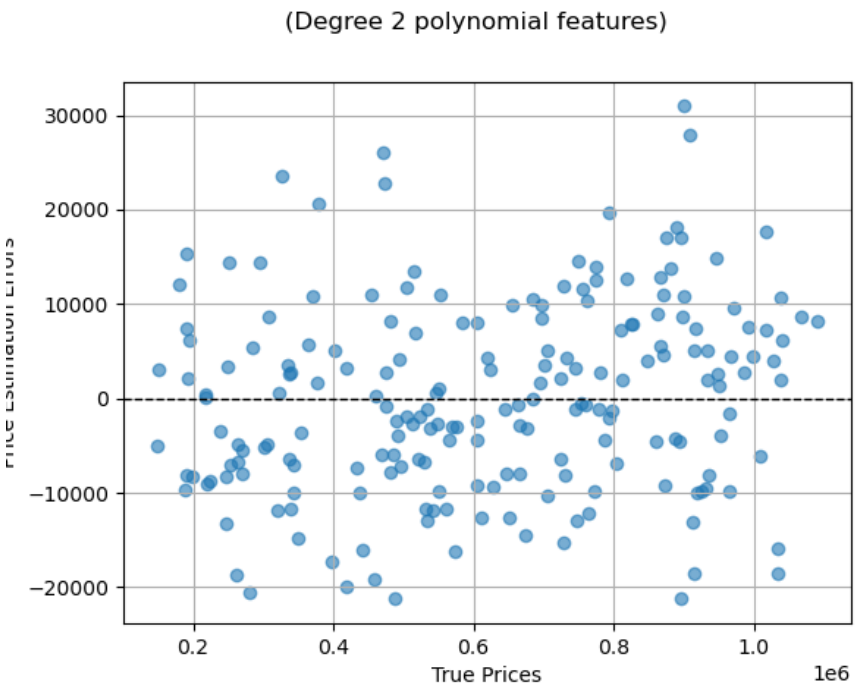
model	mse	R^2	intercept
linear	101434798.505668	0.998426	-2010567.420270
poly_deg5	138936844.541452	0.997845	-1234636.245764
poly_deg2	103781419.476088	0.998390	247486.172713
ridgePoly_deg5	1452124133.422956	0.977472	157520255.570688
ridgePoly_deg2	103544048.438783	0.998394	500815.526725

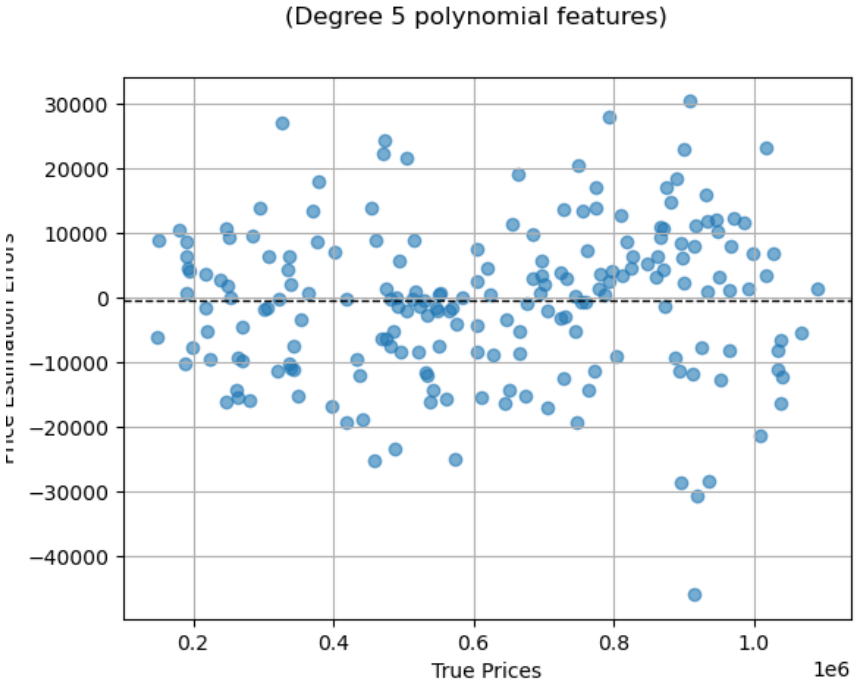
Five multiple linear regression models were fitted on the same testing and training data sets with each set consisting of the same data entities. This resulted in comparable performance. In fact, the linear model without polynomial factor enhancements and ridge regression performed the best (referred to as basic model going forward. We can see this in the R^2 and mean squared error (MSE) scores in Table 1 where there is lesser amount of cumulative error with a greater amount of variance in the data explained by the model than in the subsequent models. The seaming lack of improvement is likely due to the high level of performance that already exists in the basic model, leading to the increase in features through polynomial enhancement to result in slight overfitting. The degree of polynomial enhancement is impactful resulting in both a larger MSE score and a smaller R^2 both of which is the opposite of what is intended by polynomial enhancement. Ideally, these measures should reflect better performance as a result of the nonlinear patterns that are made more apparent. Of the two polynomial enchanced models, degree 2 performed better than degree 5 indicating that degree 5 provided too much complexity. There is a notable increase in the ridge model for polynomial degree of 5 with a magnitude increase by a factor of 10 in the MSE score, which corresponded with the lowest R^2 score.

There was significant fluctuation between intercept values. Here, the intercept has little interpretable meaning. It indicates the predicted house price values when all of the independent attributes are zero. which is outside the meaningful scope of the property characteristics considered here- for example, there are no houses you can purchase that have a nonexistent square footage and lot size. The intercept can be impactful on statistical testing for comparing linear models on the same features in different contexts, however. For example, if you were to want to compare the effect of zip code on house prices, you can compare linear models produced from the same attributes but with datasets separated by the zip code the properties were found in.

Statistically significant differences in the intercepts of corresponding linear models can indicate a meaningful difference in expected prices, even if the models have similar coefficient values. With this in mind, the intercept of output can play an important role likely meaning that analysis multiple linear models might impact what hyperparameters are used when enhancing linear models, so it seems important to consider which is best. In this case, it is assume that the basic models intercept is the most effective since it has the best performance overall.

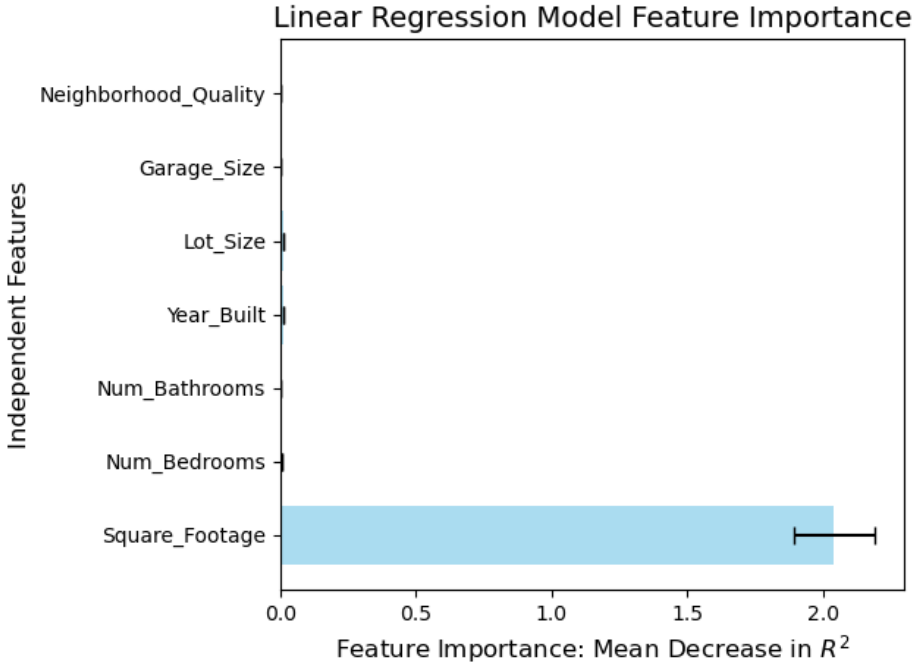




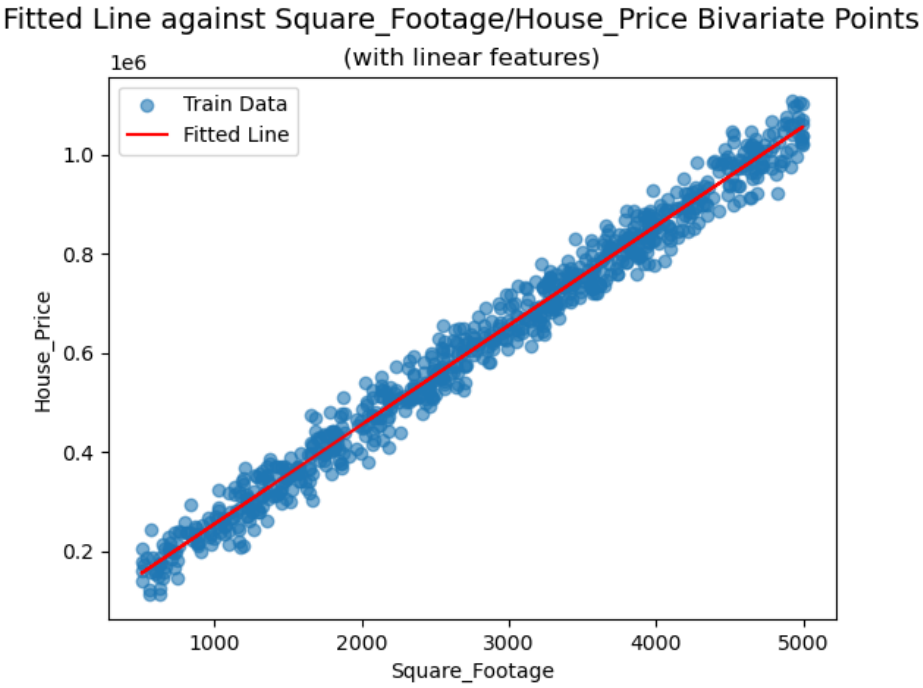


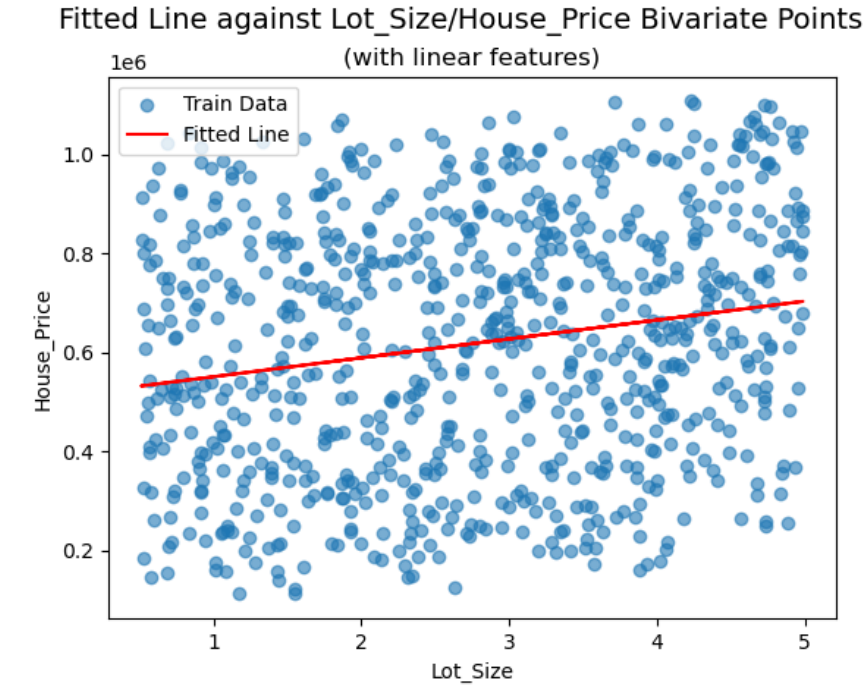
Where the MSE provides a sense of the overall precision in the model, its average can be used to clarify any bias. There is a slight definition of the residuals from the mean in the 5 degree polynomial model further indicating it having a slightly lower performance. The slight deviation persisted for the ridge model with the same polynomial features. This small deviation has little impact on the expected performance of the model, however.

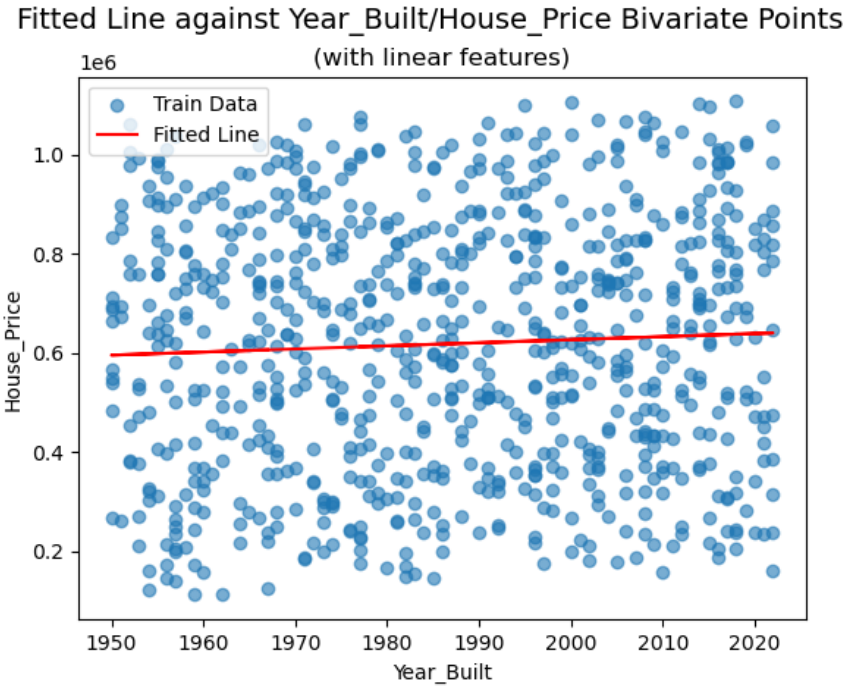
Understanding that the basic model is the best model based on the performance of the model as well as lesser computational complexity warrants further exploration of its characteristics. The individual role of each factor can be seen in the graph below.



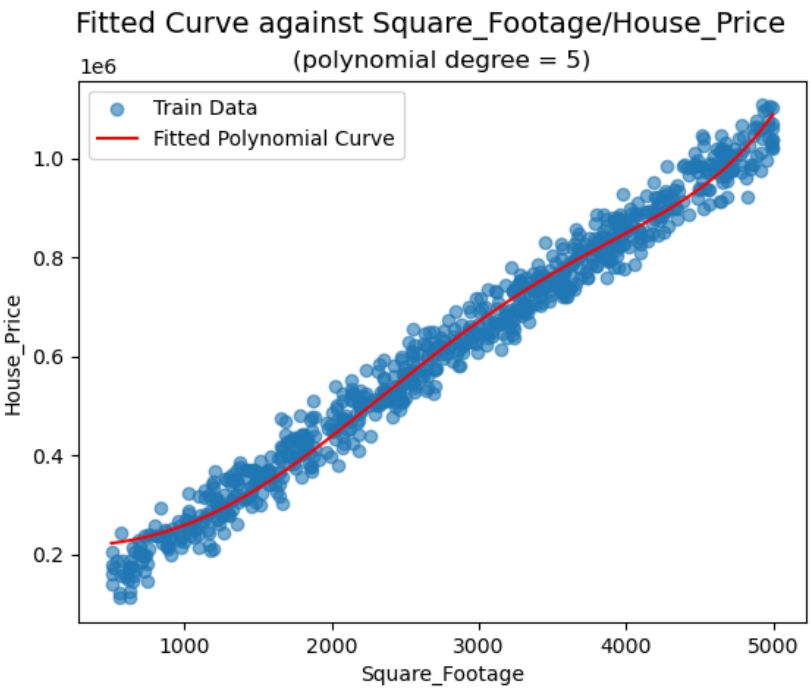
The graph shows overwhelmingly that the square footage serves as the most important feature in the model. The feature importance algorithm shuffles the values of the feature being tested while holding the other feature values in fixed positions within the data matrix and determined the effect on the R^2 value. plotting the mean decrease to the R^2 value indicates which features had the greatest role in the linear model. Here square footage had the greatest impact on the price of a house. Other impactful features included the lot size, year built and the number of bathrooms, but none of these had made a reasonable contribution overall. This is corroborated by the relationship between the fitted line of the simple regression model for square footage and the bi-variate behavior between the square footage and house price in data objects when compared to the other most important features.

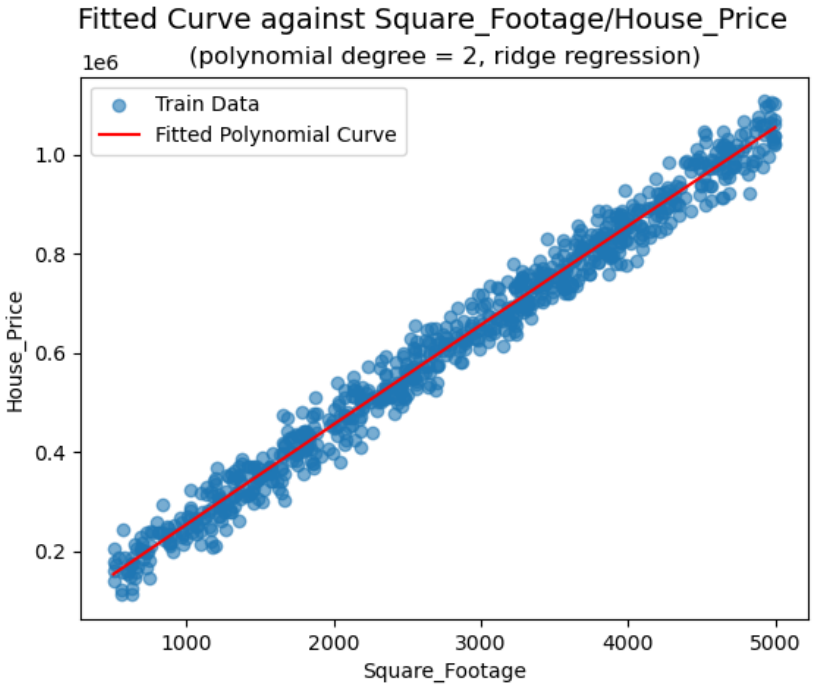

Fitted Line against Square_Footage/House_Price Bivariate Points
(with linear features)
1e6
Train Data
Fitted Line
1.0
0.8
0.6
0.4
0.2
House_Price
1000
2000
3000
4000
5000
Square_Footage


Fitted Line against Lot_Size/House_Price Bivariate Points
(with linear features)
1e6
Train Data
Fitted Line
1.0
0.8
0.6
0.4
0.2
House_Price
1
2
3
4
5
Lot_Size



This trend perpetuated across the different models as well with slightly different fitted lines reflecting the nonlinear relationships they seek to model. However, the fit line may show more change in slope in the data than actually exists.





5 Conclusion

Ultimately, the best model was the basic multiple regression linear model with the square footage feature accounting for most of the variation explained by the models created overall. The basic linear model's fit line also corresponded best with the bivariate relationship between the square footage and house price across data objects where the polynomial models suggested inflection points across the range of square footage that is not seen in the data. This shows that in some cases simpler can be better when creating models. While all models were successful, in all categories the, basic linear model served as the best option.